

Rajeev Jain

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rajeeja@gmail.com U.S. Permanent Resident

Principal Research Software Engineer | ML Infrastructure | Scientific Computing

SUMMARY

Principal research software engineer with 16+ years delivering open-source and production-grade software for exascale systems, ML infrastructure, and scientific computing. Lead experience across Python platforms, large-scale PyTorch workloads, HPC I/O optimization, reproducible experiment systems, release engineering, and multi-institution collaboration. Recent work includes scaling AI weather workflows on Aurora's Intel GPU stack, running 10,000+ benchmarking and hyperparameter optimization experiments for cancer drug response, and reducing FLASH-X checkpoint overhead by 40–70% with more than 50% storage savings.

PROFESSIONAL EXPERIENCE

Argonne National Laboratory

2009 – Present

Principal Specialist, Research Software Engineering

Chicago, IL

- Led UXarray, an open-source Python platform for analysis of unstructured climate grids, designing the core API, numerical operators, grid I/O for ESMF / MPAS / SCRIP / HEALPix, CI/CD, and release workflows used by NCAR, DOE labs, and universities.
- Built and ported AI weather workloads for Aurora's Intel GPU stack, including a PyTorch implementation of Pangu-Weather using the Spectral Fourier Neural Operator, and ran the workflow at large scale on an exascale system.
- Optimized FLASH-X checkpoint and restart paths using asynchronous HDF5 I/O, Argobots, and SZ3 / ZFP compression, reducing checkpoint overhead by 40–70% on Summit and cutting storage more than 50%.
- Architected benchmarking and hyperparameter optimization infrastructure for CANDLE / IMPROVE, executing 10,000+ training runs across Summit, Theta, and Cori and standardizing cross-study validation for a multi-lab collaboration.
- Served as principal investigator / software lead for MeshKit within DOE NEAMS, leading reactor mesh generation tools, multi-format I/O, and parallel workflows for nuclear simulation.
- Delivered release engineering, automated testing, documentation, performance debugging, and reproducible workflows across climate, AI, and simulation software used by national labs and university collaborators.
- Mentored students, research aides, and staff on scientific Python, HPC workflows, systems engineering, and open-source engineering practices.

University of Chicago

2023 – Present

Staff At-Large

Chicago, IL

- Joint appointment supporting cancer pharmacogenomics and Earth science collaborations.
- Mentor students and research staff on software design, experimentation workflows, reproducibility, and engineering practices for research codebases.

Arizona State University

2007 – 2009

Research and Teaching Assistant

Tempe, AZ

- Conducted FEM-based research on blast mitigation and design optimization while supporting structural engineering coursework.

Earlier Experience

2005 – 2007

Project Engineer, Wipro Technologies; Engineering Intern, Tata Motors

India

TECHNICAL SKILLS

Languages: Python, C++, Fortran, Bash, R, SQL

ML / AI: PyTorch, TensorFlow, NumPy, Pandas, Xarray, Scikit-learn

HPC / Systems: MPI, OpenMP, HDF5, NetCDF, MOAB, Argobots, HPC storage and I/O, exascale GPU workflows

Workflow / DevOps: GitHub Actions, Jenkins, Docker, Singularity, Parsl, Swift/T, release automation, CI/CD

Platforms: ML infrastructure, scientific data platforms, distributed experimentation, performance engineering, reproducible workflows

Domains: Climate and weather modeling, cancer pharmacogenomics, computational physics, mesh generation

SELECTED OPEN SOURCE AND PROGRAM IMPACT

- **UXarray.** Lead developer for open-source unstructured climate analysis tooling with documentation, testing, CI/CD, and community-facing releases; tutorials and presentations delivered at SC24, AMS, SciPy, EGU, and AGU.
- **CANDLE / IMPROVE.** Core contributor to cancer AI benchmarking and hyperparameter optimization infrastructure supporting a 15+ researcher collaboration across Argonne, LLNL, and ORNL; work contributed to the 2023 R&D 100 Award.
- **FLASH-X.** I/O and compression lead for a million-line multiphysics code used in astrophysics, combustion, and fluid dynamics; work contributed to the 2022 R&D 100 Award.
- **Pangu-Weather on Aurora.** PyTorch weather-model workflow adapted for Intel GPUs on Aurora, supporting DOE exascale work in Earth system modeling and forecasting.

LEADERSHIP AND SERVICE

- DOE SBIR / STTR proposal reviewer; reviewer for the *Journal of Open Research Software* and NumGrid; program committee member for NumGrid 2020.
- Panelist, “Revolutionizing Public Infrastructure: The Impact of AI and Machine Learning,” 5th Infraday Midwest Event.
- Presented technical tutorials and talks at SC24, SciPy 2023, HDF User Group 2023, AMS, EGU, and AGU.

SELECTED PUBLICATIONS

- Partin, A., ..., **Jain, R.**, et al. “Benchmarking community drug response prediction models: datasets, models, tools, and metrics for cross-dataset generalization analysis.” *Briefings in Bioinformatics*, 2025.
- **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. “Enabling Data Reduction for Flash-X Simulations.” *SC24 Workshops: DRBSD-10*, 2024.
- **Jain, R.**, Wozniak, J. M., Partin, A., et al. “Cross-HPO: Optimizing Neural Networks for Cancer Drug Response Using Hyperparameter Tuning on Multiple Pharmacogenomic Datasets.” *CAFCW at SC24*, 2024.
- Wozniak, J. M., **Jain, R.**, et al. “CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research.” *BMC Bioinformatics*, 2018.

Full publication list available on the CV and [Google Scholar](#).

AWARDS

- R&D 100 Award, CANDLE (2023).
- R&D 100 Award, FLASH-X (2022).
- ATPESC Scholar, Argonne Training Program on Extreme-Scale Computing (2015).
- Best Paper Award, International Meshing Roundtable (2010).

EDUCATION

University of Chicago	2020
M.S. in Computer Science	
Arizona State University	2009
M.S. in Structural Engineering	
Indian Institute of Technology (ISM) Dhanbad	2006
B.Tech. in Mechanical Engineering	